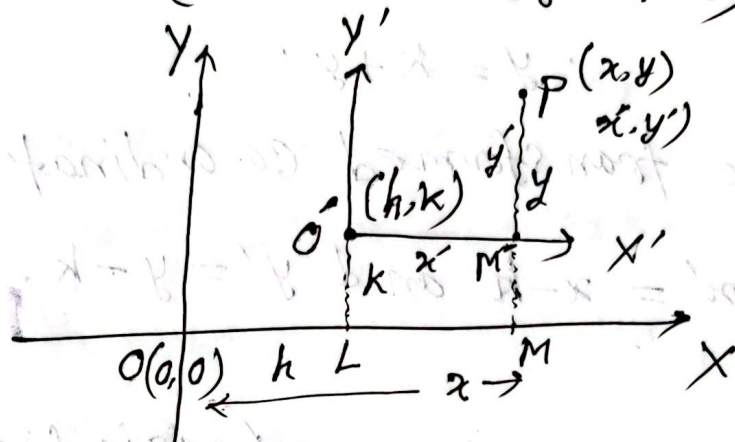


Change of axes

01521727859 } B
01836850521 }

1. Change of origin (Translation of axes) } $\begin{matrix} 01409078 \\ C \ 344 \end{matrix}$



Let us consider pair of axes $O'X'$ and $O'Y'$ parallel to the old pair OX and OY ; O' being the new origin whose co-ordinates are (α, β) referred to ~~the~~ $O'X'$ and $O'Y'$.

Let (x', y') be the co-ordinates referred to the axes $O'X'$ and $O'Y'$ or a point P whose co-ordinate referred to the old axes are (x, y) .

It is required to transform the co-ordinates (x, y) in terms of (x', y') .

From the figure, $OL = h$, $LO' = k$,

$OM = x$, $MP = y$, $O'M' = x'$ and $M'P = y'$.

Therefore, $OM = OL + LM = OL + O'M'$

$$\Rightarrow x = h + x'$$

similarly, $MP = MM' + M'P = LO' + M'P$

$$\therefore y = k + y'$$

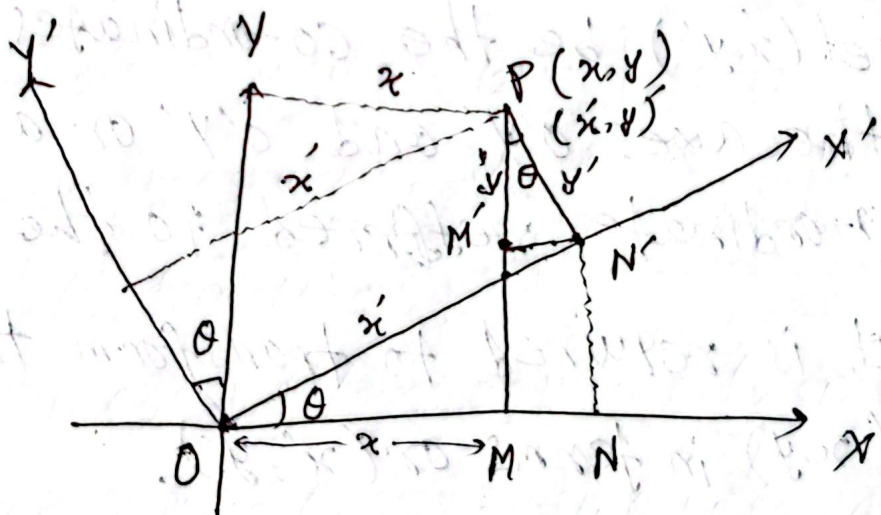
The transformed co-ordinates are

$$x' = x - k \text{ and } y' = y + k.$$

2. Q. Rotation of axes (origin fixed)

To find the change in the co-ordinates of a point when the direction of axes is turned through an angle θ where as the origin of co-ordinates remains the same.

Solution:



From the figure,

$$OM = x, \quad MP = y,$$

$$ON' = x', \quad PN' = y'.$$

Now, ~~OM~~ $OM = ON - MN$

$$= ON - M'N'$$

$$= ON' \cos \theta - PN' \sin \theta$$

$$\therefore x = x' \cos \theta - y' \sin \theta \longrightarrow (1)$$

Again, $MP = MM' + M'P$

$$= ~~OM~~ NN' + M'P$$

$$= ON' \sin \theta + PN' \cos \theta$$

$$\therefore y = x' \sin \theta + y' \cos \theta \longrightarrow (2)$$

$$(1) \times \cos \theta + (2) \times \sin \theta \Rightarrow$$

$$x \cos \theta + y \sin \theta = x' \cos^2 \theta - y' \sin \theta \cos \theta + x' \sin^2 \theta$$

$$+ y' \sin \theta \cos \theta$$

$$= x' (\cos^2 \theta + \sin^2 \theta)$$

$$= x'$$

$$\therefore x' = x \cos \theta + y \sin \theta$$

$$(1) \times -\sin \theta + (2) \times \cos \theta \Rightarrow$$

$$-x \sin \theta + y \cos \theta = -x' \sin \theta \cos \theta + y' \sin^2 \theta$$

$$+ x' \sin \theta \cos \theta + y' \cos^2 \theta$$

$$\Rightarrow -x \sin \theta + y \cos \theta = y' (\sin^2 \theta + \cos^2 \theta)$$

$$= y'$$

$$\therefore y' = \frac{x \cos \theta + y \sin \theta}{-x \sin \theta + y \cos \theta}$$

$$x = x' \cos \theta - y' \sin \theta$$

$$y = x' \sin \theta + y' \cos \theta$$

$$x' = x \cos \theta + y \sin \theta$$

$$y' = -x \sin \theta + y \cos \theta$$

	x'	y'
x	$\cos \theta$	$-\sin \theta$
y	$\sin \theta$	$\cos \theta$

3. Q. Find the angle θ through which the axes are to be rotated so as to remove the xy term in the equation $x^2 + 4xy + y^2 - 2x + 2y - 6 = 0$

Solution: Given that, $x^2 + 4xy + y^2 - 2x + 2y - 6 = 0 \rightarrow ($

We know,

$$x = x' \cos \theta - y' \sin \theta \rightarrow (2)$$

$$y = x' \sin \theta + y' \cos \theta \rightarrow (3)$$

	x'	y'
x	$\cos \theta$	$-\sin \theta$
y	$\sin \theta$	$\cos \theta$

Using (2) and (3), from (1)

$$\begin{aligned} & (x' \cos \theta - y' \sin \theta)^2 + 4(x' \cos \theta - y' \sin \theta)(x' \sin \theta + y' \cos \theta) \\ & + (x' \sin \theta + y' \cos \theta)^2 - 2(x' \cos \theta - y' \sin \theta) + 2(x' \sin \theta + y' \cos \theta) \\ & - 6 = 0 \end{aligned}$$

Remove xy term;

$$\begin{aligned} & -2x'y' \cos \theta \sin \theta + 4x'y' \cos^2 \theta - 4x'y' \sin^2 \theta \\ & + 2x'y' \sin \theta \cos \theta = 0 \end{aligned}$$

$$\Rightarrow \cos^2 \theta - \sin^2 \theta = 0$$

$$\Rightarrow \cos 2\theta = 0$$

$$\Rightarrow \cos 2\theta = \cos \frac{\pi}{2}$$

$$\Rightarrow 2\theta = \frac{\pi}{2}$$

$$\therefore \theta = \frac{\pi}{4} \underline{\underline{A}}$$

4.Q. When axes are rotated through an angle 45° , the transformed equation of a curve is $17x'^2 - 16x'y' + 17y'^2 = 225$. Find the original equation of the curve.

Solution: Given that, the transformed equation of a curve is $17x'^2 - 16x'y' + 17y'^2 = 225 \rightarrow (1)$

$$x' = x \cos \theta + y \sin \theta$$

$$= x \cos 45^\circ + y \sin 45^\circ$$

$$= x \cdot \frac{1}{\sqrt{2}} + y \cdot \frac{1}{\sqrt{2}}$$

$$= \frac{x+y}{\sqrt{2}} \rightarrow (2)$$

	x'	y'
x	$\cos \theta$	$-\sin \theta$
y	$\sin \theta$	$\cos \theta$

$$y' = -x \sin \theta + y \cos \theta = -x \sin 45^\circ + y \cos 45^\circ$$

$$= -x \cdot \frac{1}{\sqrt{2}} + y \cdot \frac{1}{\sqrt{2}}$$

$$= \frac{-x+y}{\sqrt{2}} \rightarrow (3)$$

Using (2) and (3), from (1)

$$17. \left(\frac{x+y}{\sqrt{2}}\right)^2 - 16 \left(\frac{x+y}{\sqrt{2}}\right) \cdot \left(\frac{-x+y}{\sqrt{2}}\right) + 17 \left(\frac{-x+y}{2}\right)^2 = 225$$

$$\Rightarrow \frac{17}{2} (x+y)^2 - \frac{16}{2} (-x^2+y^2) + \frac{17}{2} (-x+y)^2 = 225$$

$$\Rightarrow \frac{17}{2} \cdot \{2(x^2+y^2)\} - 8(-x^2+y^2) = 225$$

$$\Rightarrow 25x^2 + 9y^2 = 225. \underline{\underline{\text{Ans.}}}$$

5. Q: If by the rotation of the rectangular co-ordinate axes about the origin, the expression $ax^2 + 2hxy + by^2$ changes to $a'x'^2 + 2h'x'y' + b'y'^2$, $a+b = a'+b'$ and $ab-h^2 = a'b'-h'^2$.

Solution: Let (x, y) be the co-ordinates of a point P referred to a set of rectangular axes. If the axes are rotated through an angle θ about the origin, let the co-ordinates of the same point P be (x', y') referred to the new system of rectangular axes. Then we have by

$$\therefore x = x' \cos \theta - y' \sin \theta$$

$$y = x' \sin \theta + y' \cos \theta$$

	x'	y'
x	$\cos \theta$	$-\sin \theta$
y	$\sin \theta$	$\cos \theta$

Now, $ax^2 + 2hxy + by^2$ becomes

$$a(x'\cos\theta - y'\sin\theta)^2 + 2h(x'\cos\theta - y'\sin\theta)(x'\sin\theta + y'\cos\theta) + b(x'\sin\theta + y'\cos\theta)^2$$

$$= a(x'^2\cos^2\theta - 2x'y'\cos\theta\sin\theta + y'^2\sin^2\theta)$$

$$+ 2h\{x'^2\sin\theta\cos\theta + x'y'\cos^2\theta - x'y'\sin^2\theta - y'^2\sin\theta\cos\theta\}$$

$$+ b(x'^2\sin^2\theta + 2x'y'\sin\theta\cos\theta + y'^2\cos^2\theta)$$

$$= (a\cos^2\theta + 2h\sin\theta\cos\theta + b\sin^2\theta)x'^2$$

$$+ 2\{-a\cos\theta\sin\theta + h(\cos^2\theta - \sin^2\theta) + b\sin\theta\cos\theta\}x'y'$$

$$+ (a\sin^2\theta - 2h\sin\theta\cos\theta + b\cos^2\theta)y'^2$$

$$= a'x'^2 + 2h'x'y' + b'y'^2$$

$$\text{Where, } a' = a\cos^2\theta + 2h\sin\theta\cos\theta + b\sin^2\theta \rightarrow (1)$$

$$h' = -a\cos\theta\sin\theta + h(\cos^2\theta - \sin^2\theta) + b\sin\theta\cos\theta \rightarrow (2)$$

$$b' = a\sin^2\theta - 2h\sin\theta\cos\theta + b\cos^2\theta \rightarrow (3)$$

$$(1)+(2) \Rightarrow a' + b' = a + b$$

Now, $2a' = 2a \cos 2\theta + 4h \sin \theta \cos \theta + 2b \sin 2\theta$

$$= a(1 + \cos 2\theta) + 2h \sin 2\theta + b(1 - \cos 2\theta)$$

$$= (a+b) + \{2h \sin 2\theta + (a-b) \cos 2\theta\}$$

Similarly, $2b' = (a+b) - \{2h \sin 2\theta + (a-b) \cos 2\theta\}$

and

$$2h' = -2a \sin \theta \cos \theta + 2h \cos 2\theta + 2b \sin \theta \cos \theta$$

$$= 2h \cos 2\theta - (a-b) \sin 2\theta$$

$$\therefore \cancel{(2h')}^r = 4h'^r = 4h^r \cos 2\theta - 4h(a-b) \sin 2\theta \cos 2\theta + (a-b)^r \sin 2\theta$$

$$\therefore 4a'b' = (a+b)^r + \{2h \sin 2\theta + (a-b) \cos 2\theta\}^r$$

$$= (a+b)^r - 4h^r \sin^2 2\theta - 4h(a-b) \sin 2\theta \cos 2\theta - (a-b)^r \cos^2 2\theta$$

$$\therefore 4(a'b' - h'^r) = (a+b)^r - 4h^r - (a-b)^r$$

$$= 4ab - 4h^r$$

$$\therefore a'b' - h'^r = ab - h^r \quad \text{(\underline{proved})}$$

6. Q. Determine the equation of the curve $2x^2 + 3y^2 - 8x + 6y - 7 = 0$ when the origin is transferred to the point $(2, -1)$.

Solution: When the origin is transferred to the point $(2, -1)$ put $x = x' + 2$, $y = y' - 1$ in the above equation. Then $2(x' + 2)^2 + 3(y' - 1)^2 - 8(x' + 2) + 6(y' - 1) - 7 = 0$

$$\Rightarrow 2x'^2 + 3y'^2 = 18. \text{ Ans. } \text{ or, } 2x^2 + 3y^2 = 18$$

Removing subscripts

7. Q. Determine the equation of the parabola $x^2 - 2xy + y^2 + 2x - 4y + 3 = 0$ after rotating of axes through 45° .

Solution: When the axes have been rotated through an angle 45° , then

$$x = x' \cos \theta - y' \sin \theta$$

$$= x' \cos 45^\circ - y' \sin 45^\circ = \frac{x' - y'}{\sqrt{2}}$$

$$y = x' \sin 45^\circ + y' \cos 45^\circ = \frac{x' + y'}{\sqrt{2}}$$

$$\therefore \left(\frac{x' - y'}{\sqrt{2}}\right)^2 - 2\left(\frac{x' - y'}{\sqrt{2}}\right)\left(\frac{x' + y'}{\sqrt{2}}\right) + \left(\frac{x' + y'}{\sqrt{2}}\right)^2 + 2\left(\frac{x' - y'}{\sqrt{2}}\right) - 4\left(\frac{x' + y'}{\sqrt{2}}\right) + 3 = 0.$$

	x'	y'
x	$\cos \theta$	$-\sin \theta$
y	$\sin \theta$	$\cos \theta$

on simplification the equation reduces to

$$2y'^2 - \sqrt{2}x' - 3\sqrt{2}y' + 3 = 0.$$

or, $2y^2 - \sqrt{2}x - 3\sqrt{2}y + 3 = 0$ [removing suffixes]

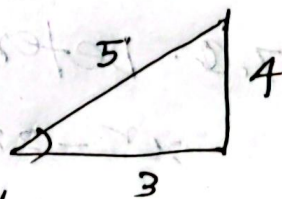
or, Determine the transformed equation of ... of which the origin remain unchanged and the direction of axes is turned through an angle $\tan^{-1}(-\frac{4}{3})$.

8. Q. Determine the transformed equation of the $11x^2 + 24xy + 4y^2 - 20x - 40y - 5 = 0$ equation when the axes are rotated by an angle of $\tan^{-1}(-\frac{4}{3})$ keeping the origin fixed.

Solution: Given that, $\theta = \tan^{-1}(-\frac{4}{3})$

$$\Rightarrow \tan \theta = -\frac{4}{3}$$

Now, $\sin \theta = -\frac{4}{5}$, $\cos \theta = \frac{3}{5}$



$\therefore$ The transformed equation of the

$11x^2 + 24xy + 4y^2 - 20x - 40y - 5 = 0$ equation is

$$11(x'\cos\theta - y'\sin\theta)^2 + 24(x'\cos\theta - y'\sin\theta)(x'\sin\theta + y'\cos\theta) + 4(x'\sin\theta + y'\cos\theta)^2$$

	x'	y'
x	$\cos\theta$	$-\sin\theta$
y	$\sin\theta$	$\cos\theta$

$$- 20(x'\cos\theta - y'\sin\theta) - 40(x'\sin\theta + y'\cos\theta) - 5 = 0$$

$$\Rightarrow 11\left(x' \cdot \frac{3}{5} - y' \cdot \frac{4}{5}\right)^2 + 24\left(x' \cdot \frac{3}{5} - y' \cdot \frac{4}{5}\right)\left(x' \cdot \frac{4}{5} + y' \cdot \frac{3}{5}\right)$$

$$+ 4\left(x' \cdot \frac{4}{5} + y' \cdot \frac{3}{5}\right)^2 - 20\left(x' \cdot \frac{3}{5} - y' \cdot \frac{4}{5}\right)$$

$$- 40\left(x' \cdot \frac{4}{5} + y' \cdot \frac{3}{5}\right) - 5 = 0 \Rightarrow x^2 - 4y^2 - 4x + 8y + 12 = 0$$

9. Q. Transform the equation $11x^2 + 24xy + 4y^2 - 20x - 40y - 5 = 0$ to rectangular axes through the point $(2, -1)$ and inclined at an angle $\tan^{-1} \frac{4}{3}$.

10. Q. Show that if the axes are rotated through an angle $\theta = \frac{1}{2} \tan^{-1} \frac{2h}{a-b}$, then the xy term from the expression $ax^2 + 2hxy + by^2$ vanishes.

Solution: From 5. Q. [5. Q. থেকে আগের ব্যাকটাইক নিখাও হবে]

$$2h' = 2h \cos 2\theta - (a-b) \sin 2\theta$$

In order to remove the $x'y'$ term, the Coefficient h' must be zero.

$$\therefore 0 = 2h \cos 2\theta - (a-b) \sin 2\theta$$

$$\Rightarrow (a-b) \sin 2\theta = 2h \cos 2\theta$$

$$\Rightarrow \tan 2\theta = \frac{2h}{a-b}$$

$$\Rightarrow 2\theta = \tan^{-1} \frac{2h}{a-b}$$

$$\therefore \theta = \frac{1}{2} \tan^{-1} \frac{2h}{a-b} \quad \text{(shown)}$$

11.a. Determine the angle through which the axes must be rotated to remove the xy term in the equation $7x^2 - 6\sqrt{3}xy + 13y^2 = 16$. Also find the transformed equation.

Solution: The given equation is

$$7x^2 - 6\sqrt{3}xy + 13y^2 = 16 \longrightarrow (1)$$

Compare the equation (1) with $ax^2 + 2hxy + by^2$

we get $a = 7, h = -3\sqrt{3}, b = 13$

We know that if the axes are rotated through an angle $\theta = \frac{1}{2} \tan^{-1} \frac{2h}{a-b}$, then the xy term from the expression $ax^2 + 2hxy + by^2$ vanishes

$$\therefore \theta = \frac{1}{2} \tan^{-1} \frac{2h}{a-b}$$

$$= \frac{1}{2} \tan^{-1} \frac{2(-3\sqrt{3})}{7-13} = \frac{1}{2} \tan^{-1} (\sqrt{3}) = 30^\circ$$

When the axes have been rotated through an angle 30° , then

$$x = x' \cos \theta - y' \sin \theta$$

$$= x' \cos 30^\circ - y' \sin 30^\circ = x' \cdot \frac{\sqrt{3}}{2} - y' \cdot \frac{1}{2} = \frac{\sqrt{3}x' - y'}{2}$$

and, $y = x' \sin \theta + y' \cos \theta = x' \sin 30^\circ + y' \cos 30^\circ$

$$= x' \cdot \frac{1}{2} + y' \cdot \frac{\sqrt{3}}{2} = \frac{x' + \sqrt{3}y'}{2}$$

	x'	y'
x	$\cos \theta$	$-\sin \theta$
y	$\sin \theta$	$\cos \theta$

Putting these values in (1), we get

$$7 \left(\frac{\sqrt{3}x' - y'}{2} \right)^2 - 6\sqrt{3} \cdot \left(\frac{\sqrt{3}x' - y'}{2} \right) \left(\frac{x' + \sqrt{3}y'}{2} \right) + 13 \left(\frac{x' + \sqrt{3}y'}{2} \right)^2 = 16$$

$$\Rightarrow \frac{7}{4} (3x'^2 - 2\sqrt{3}x'y' + y'^2) - \frac{3\sqrt{3}}{2} (\sqrt{3}x'^2 + 2x'y' + \sqrt{3}y'^2) + \frac{13}{4} (x'^2 + 2\sqrt{3}x'y' + 3y'^2) = 16$$

$$\Rightarrow 4x'^2 + 16y'^2 = 16$$

$$\therefore x'^2 + 4y'^2 = 4$$

Now dropping the suffixes, the transformed equation is $x^2 + 4y^2 = 4$.

12. Q. Transform the equation $x^2 + 2\sqrt{3}xy - y^2 = 2a^2$ the axes inclined at an angle 30° to the original axes.

13. Q. If by the rotation of the rectangular co-ordinate axes about the origin the expression $ax + by$ changes to $a'x' + b'y'$, prove that $a'^2 + b'^2 = a^2 + b^2$.

Change of axes

Mominul Islam
Lecturer
Department of Mathematics
Rajshahi University of Engineering & Technology
RUET, Rajshahi-6204, Bangladesh

* Removal of first degree terms: Shifting the origin to the desired point, transform the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ to the form $ax^2 + 2hxy + by^2 + c_1 = 0$, where the direction of axes remains unaltered.

Solution: The given equation is,

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \rightarrow (1)$$

Let the origin be shifted to the point (α, β) without changing the direction of the axes. The equation (1) can be written as

$$a(x+\alpha)^2 + 2h(x+\alpha)(y+\beta) + b(y+\beta)^2 + 2g(x+\alpha) + 2f(y+\beta) + c = 0$$

$$\Rightarrow ax^2 + 2hxy + by^2 + (a\alpha + h\beta + g) \cdot 2x + (h\alpha + b\beta + f) \cdot 2y + (a\alpha^2 + 2h\alpha\beta + b\beta^2 + 2g\alpha + 2f\beta + c) = 0 \rightarrow (2)$$

When remove the first degree terms, then coefficients of x and y is zero.

$$\therefore a\alpha + h\beta + g = 0$$

$$h\alpha + b\beta + f = 0$$

solving these equation.

$$(\alpha, \beta) = \left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2} \right)$$

$$\text{Now, } a\alpha^2 + 2h\alpha\beta + b\beta^2 + 2g\alpha + 2f\beta + C$$

$$= a\alpha^2 + h\alpha\beta + g\alpha + \cancel{h\alpha\beta} + b\beta^2 + f\beta + g\alpha + f\beta + C$$

$$= \alpha(a\alpha + h\beta + g) + \beta(h\alpha + b\beta + f) + g\alpha + f\beta + C$$

$$= g\alpha + f\beta + C$$

$$= C_1$$

putting these values in (2), we get

$$ax^2 + 2hxy + by^2 + C_1 = 0.$$

$$\text{or, Let } F(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + C = 0 \rightarrow (1)$$

$$\therefore \frac{\partial F}{\partial x} = 2ax + 2hy + 2g = 0$$

$$\Rightarrow ax + hy + g = 0 \rightarrow (2)$$

$$\& \frac{\partial F}{\partial y} = 2hx + 2by + 2f = 0$$

$$\Rightarrow hx + by + f = 0 \rightarrow (3)$$

Solve the equations (2) & (3), we get

$$x = \frac{hf - bg}{ab - h^2}, y = \frac{gh - af}{ab - h^2}$$

$$\text{or, } \alpha = \frac{hf - bg}{ab - h^2}, \beta = \frac{gh - af}{ab - h^2}$$

shifting the origin to the point (x_1, y_1) the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + C = 0$ takes the form $ax^2 + 2hxy + by^2 + C_1 = 0$, where $C_1 = gd + fb + C$.

Problem-1: Remove the first degree terms of the equation $36x^2 + 24xy + 29y^2 - 72x + 126y + 81 = 0$.

Solution: The given equation is,

$$36x^2 + 24xy + 29y^2 - 72x + 126y + 81 = 0 \rightarrow (1)$$

$$\text{Let, } F(x, y) = 36x^2 + 24xy + 29y^2 - 72x + 126y + 81 = 0$$

$$\therefore \frac{\partial F}{\partial x} = 72x + 24y - 72 = 0$$

$$\Rightarrow 3x + y - 3 = 0 \rightarrow (2)$$

$$\& \frac{\partial F}{\partial y} = 24x + 58y + 126 = 0$$

$$\Rightarrow 12x + 29y + 63 = 0 \rightarrow (3)$$

Solve equations (2) & (3), we get

$$\therefore (x, y) = (2, -3) = (\alpha, \beta)$$

To remove the first degree term, shifting the origin to the point $(2, -3)$, we get from (1) $\Rightarrow$

$$36x^2 + 24xy + 29y^2 + C_1 = 0 \rightarrow (4)$$

Where, $G_1 = g\alpha + f\beta + C$

$$= -36.2 + 63.(-3) + 81$$

$$= -180$$

putting the value of G_1 in equation (4), we get

$$36x^2 + 24xy + 29y^2 - 180 = 0.$$

Problem-2: Transform the equation $11x^2 - 4xy + 14y^2 - 58x - 44y + 126 = 0$ to new axes of x and y whose equations are the mutually perpendicular lines $x - 2y + 1 = 0$ and $2x + y - 8 = 0$ respectively.

Solution: The point of intersection of the two lines $x - 2y + 1 = 0$ and $2x + y - 8 = 0$ is

$$\frac{x}{15} = \frac{y}{10} = \frac{1}{5}$$

$$\therefore (x, y) = (3, 2)$$

Therefore, the point of intersection is $(3, 2)$.

The given equation is,

$$11x^2 - 4xy + 14y^2 - 58x - 44y + 126 = 0 \rightarrow (1)$$

Shift the origin to the point $(3, 2)$, from (1)

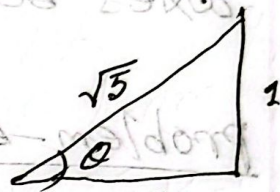
$$11(x+3)^2 - 4(x+3)(y+2) + 14(y+2)^2 - 58(x+3) - 44(y+2) + 126 = 0$$

$$\Rightarrow 11x^2 - 4xy + 14y^2 - 5 = 0 \rightarrow (2)$$

If the rotate axes through an angle θ , then the equation $x - 2y + 1 = 0$ makes angle θ with x -axis.

$$\therefore \tan \theta = \frac{1}{2}$$

$$\Rightarrow \sin \theta = \frac{1}{\sqrt{5}}, \cos \theta = \frac{2}{\sqrt{5}}$$



When the axes have been rotated through an angle θ , then

$$x = x' \cos \theta - y' \sin \theta = x' \cdot \frac{2}{\sqrt{5}} - y' \cdot \frac{1}{\sqrt{5}} = \frac{2x' - y'}{\sqrt{5}}$$

$$y = x' \sin \theta + y' \cos \theta$$

$$= x' \cdot \frac{1}{\sqrt{5}} + y' \cdot \frac{2}{\sqrt{5}} = \frac{x' + 2y'}{\sqrt{5}}$$

	x'	y'
x	$\cos \theta$	$-\sin \theta$
y	$\sin \theta$	$\cos \theta$

Putting these values in (2), we get

$$11 \left(\frac{2x' - y'}{\sqrt{5}} \right)^2 - 4 \left(\frac{2x' - y'}{\sqrt{5}} \right) \left(\frac{x' + 2y'}{\sqrt{5}} \right) + 14 \left(\frac{x' + 2y'}{\sqrt{5}} \right)^2 - 5 = 0$$

$$\Rightarrow \frac{11}{5} (4x'^2 - 4x'y' + y'^2) - \frac{4}{5} (2x'^2 + 4x'y' - x'y' - 2y'^2) + \frac{14}{5} (x'^2 + 4x'y' + 4y'^2) - 5 = 0$$

$$\Rightarrow (44 - 8 + 14)x'^2 + (11 + 8 + 56)y'^2 - 25 = 0$$

$$\Rightarrow 2x'^2 + 3y'^2 - 1 = 0 \quad \text{or, } 2x'^2 + 3y'^2 = 1. \quad \underline{\underline{\text{Ans.}}}$$

Problem-3: Transform the equation $17x^2 + 18xy - 7y^2 - 16x - 32y - 18 = 0$ to one in which there is no term involving x, y and xy , both sets of axes being rectangular.

Problem-4: The equation $3x^2 + 2xy + 3y^2 - 18x - 22y + 50 = 0$ is transformed to $4x'^2 + 2y'^2 = 1$ when referred to rectangular axes through the point $(2, 3)$. Find the inclination of the later axes to the former.

Problem-5: Find the transformation equation of $3x^2 + 2xy + 3y^2 - 18x - 22y + 50 = 0$ when origin is shifted at $(2, 3)$ and the axes is turned through an angle 45° .